1. Convex shape

1. Convex sets and functions

$C \subseteq \mathbb{R}^d$ is convex if, for all $x, y \in C$ and $\lambda \in [0,1]$,

$$\lambda x + (1-\lambda)y \in C.$$

On a convex C, $f : C \to \mathbb{R}$ is convex if

$$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda)f(y).$$

Prove convexity for arbitrary x, y, λ. Refute it with one escaping segment or one graph point above its chord. A convex feasible set and a convex objective are separate conditions. Every norm ball $\{w : \|w\|_p \le r\}$ with $p \ge 1$, $r \ge 0$ is convex.

2. Strict convexity and three distinct claims

Strict convexity makes the chord inequality strict for $x \ne y$ and $0 < \lambda < 1$. If a minimizer exists, it is unique. It does not prove that a minimum is attained.

- **Existence:** is the infimum attained?
- **Global optimality:** does a candidate beat every feasible point?
- **Uniqueness:** can two different minimizers tie?

A convex function can fail existence, and a convex function can have many minimizers. Keep all three claims separate.

2. First-order certificates

1. Differentiable convex objective

Let f be differentiable on an open convex set C. Then f is convex iff

$$f(y) \ge f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y \in C.$$

Every tangent hyperplane is therefore a global lower bound. In the unconstrained case,

$$\nabla f(x^*) = 0 \implies x^* \text{ is a global minimizer.}$$

The conclusion uses convexity; a zero gradient alone is not enough for a nonconvex function.

2. Nonsmooth and constrained cases

For convex $f : \mathbb{R}^d \to \mathbb{R}$, g is a subgradient at x if

$$f(y) \ge f(x) + g^T (y - x) \quad \text{for every } y \in \mathbb{R}^d.$$

$\partial f(x)$ is the set of all subgradients. Thus $0 \in \partial f(x^*)$ certifies global optimality. For $f(x) = |x|$, $\partial f(0) = [-1, 1]$.

If C is convex and f is differentiable and convex, feasible x^* minimizes f over C iff

$$\nabla f(x^*)^T (y - x^*) \ge 0 \quad \text{for every } y \in C.$$

A boundary optimum need not have zero gradient.

3. Select the legal certificate

- Differentiable convex, unconstrained: $\nabla f(x^*) = 0$.
- Convex and possibly nonsmooth: $0 \in \partial f(x^*)$.
- Differentiable convex over convex C: every feasible displacement has nonnegative gradient inner product.
- Strictly convex plus existence: any valid global certificate identifies the unique minimizer.

The first three certificates imply global optimality, not uniqueness by themselves.

3. Curvature and descent

1. Hessian certificates

A symmetric H is PSD ($H \succeq 0$) if $z^T H z \ge 0$ for every z; it is PD ($H \succ 0$) if the inequality is strict for every nonzero z.

If f is twice continuously differentiable on an open convex domain,

$$f \text{ is convex iff } \nabla^2 f(x) \succeq 0 \text{ for every } x.$$

$\nabla^2 f(x) \succ 0$ everywhere is sufficient for strict convexity, not necessary.

For $\lambda > 0$,

$$J(w) = \|Aw - b\|_2^2 + \lambda \|w\|_2^2,$$

$$z^T (A^T A + \lambda I) z = \|Az\|_2^2 + \lambda \|z\|_2^2 > 0$$

for nonzero z, so J is strictly convex even when A has dependent columns.

2. L-smoothness and the descent lemma

f is L-smooth in the Euclidean norm if

$$\|\nabla f(x) - \nabla f(y)\|_2 \le L \|x - y\|_2.$$

Then

$$f(y) \le f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} \|y - x\|_2^2.$$

For $y = x - \eta \nabla f(x)$,

$$f(y) \le f(x) - \eta \Big(1 - L\frac{\eta}{2}\Big) \|\nabla f(x)\|_2^2.$$

Hence $0 < \eta \le \frac{1}{L}$ is a simple sufficient non-increase range, with strict decrease when the gradient is nonzero. This is a one-step guarantee, not a complete rate theorem. Smoothness is an upper-curvature bound; it is not strict or strong convexity.

3. Gradient method and step-size diagnosis

$$w_{t+1} = w_t - \eta \nabla f(w_t), \quad \eta > 0.$$

The update is not by itself a convergence theorem. For $f(w) = \frac{1}{2}(w - 3)^2$ and $e_t = w_t - 3$,

$$e_{t+1} = (1 - \eta) e_t.$$

- $\eta = \frac{1}{2}$: same side, distance halves;
- $\eta = \frac{3}{2}$: crosses the minimizer, distance halves;
- $\eta = 2$: alternates; objective does not decrease;
- $\eta = \frac{5}{2}$: alternates with growing distance.

The displayed descent range is sufficient, not a characterization of every convergent step.

4. Computational lookup

1. Big-O, encoding, polynomial time

- $f(n) = O(g(n))$ iff some $c > 0, n_0$ satisfy $f(n) \le cg(n)$ for every $n \ge n_0$. Constants are independent of every declared scaling variable.
- $T(n, d) = 3nd^2 + 7d^3 + 50$ is $O(nd^2 + d^3)$ when n, d scale, but $O(n)$ when d is fixed.
- Polynomial time means at most cN^k operations in encoded input length N, under a fixed model and encoding.
- If binary M has length $N \approx \log_2 M$, an M-step loop is exponential in N. Enumerating $\{0, 1\}^d$ takes 2^d evaluations.
- Big-O is an eventual upper bound, not an equality or matching lower bound; every fixed-degree polynomial is eventually dominated by 2^n.

2. Decision problems, P, and NP

- **Decision problem:** one yes/no answer per encoded instance.
- $\mathbf{P}$**:** a deterministic algorithm decides both yes- and no-instances in polynomial time.
- $\mathbf{NP}$**:** some polynomial q and polynomial-time verifier V satisfy

$$x \in L \leftrightarrow \exists c : |c| \le q(|x|) \quad \text{and} \quad V(x, c) = 1.$$

Every yes-instance has a polynomial-length accepting certificate; no certificate is accepted for a no-instance. Checking a supplied certificate is not finding one. A P solver can ignore c and serve as a verifier, so $\mathrm{P} \subseteq \mathrm{NP}$; whether $\mathrm{P} = \mathrm{NP}$ is open.

3. Reduction direction and NP-hardness

$A \to B$ means a polynomial-time map r preserves yes/no answers:

$$x \in A \leftrightarrow r(x) \in B.$$

- A fast solver for the **target** B gives a fast solver for the **source** A; hence B is at least as hard as A.
- B is NP-hard if every problem in NP reduces to B; B need not be in NP.
- If known NP-hard A reduces to B, then B is NP-hard; a polynomial-time exact solver for B would imply $\mathrm{P} = \mathrm{NP}$.
- For an optimization target, name its decision version or show that an **exact optimizer** would decide an NP-hard decision problem. The class NP applies here to decision problems, not directly to that optimization target.

4. Common traps and the three audits

- One 2^d implementation proves only that this algorithm is exponential. It proves neither NP-hardness nor an exponential lower bound.
- It also does not rule out a faster algorithm, approximation, or heuristic.
- Reversing a reduction reverses its implication.
- **Statistical:** do assumptions support accuracy?
- **Optimization:** do geometry and curvature license the certificate or step?
- **Computational:** is runtime polynomial in every encoded input?

Do not infer computation from Hoeffding, data sufficiency from convexity, or statistical accuracy from efficient runtime. A computational–statistical gap requires sample sufficiency and a computational obstruction for the same task under the same assumptions.